Eskandar (Alex) Seif

Principal Software Architect | Public Sector & Institutional Systems Lead

Specialization: High-Concurrency Distributed Platforms | Public Sector & Regulated Systems (Saudi SASO, SAMA, Egyptian MoI) | Donor Transparency & NGO Digital Infrastructure (UN, GIZ, EU, AfDB) | Cloud & Database Architecture

- **Location:** Cairo, Egypt (Available for Immediate Remote / B2B Contract | Open to Relocation)
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PROFESSIONAL SUMMARY

Principal Software Architect and Institutional Technology Consultant with over 20 years of hands-on experience designing and delivering mission-critical digital infrastructure for federal ministries, public sector authorities, multilateral donor programs (UN, GIZ, EU, AfDB), and regulated fintech. Demonstrated track record architecting nationwide vehicle inspection and booking ecosystems (Saudi SASO, Aman Portal), air-gapped security platforms for law enforcement, PCI DSS compliant payment gateways, and zero-downtime enterprise cloud migrations.

Pragmatic mastery across distributed systems, decoupled API architectures, high-concurrency state machines, 3NF/Star Schema relational optimization, and deterministic CI/CD automation. Focuses on robust software architecture, GDPR compliance, and verifiable data integrity.

TECHNICAL STACK & ARCHITECTURE COMPETENCIES

- **Enterprise Architecture & System Design:** Distributed Systems, Decoupled Architecture, High-Concurrency Routing, Transactional State Machines, Domain-Driven Design (DDD), REST & OpenAPI Contract Synchronization, Microservices & Containerization.
- **Public Sector & Institutional Systems:** Federal Regulatory Systems, Air-Gapped Deployments, Multi-Tier Environment Governance (Dev/Test/Stage/Pre-Prod), Audit Trail Preservation, Donor Transparency Platforms.
- **Cloud & DevOps Infrastructure:** AWS (ECS, Aurora PostgreSQL, S3, CloudFront, Route 53), Infrastructure as Code (Terraform), Docker / Docker Compose, Linux Administration (Debian, CentOS, Oracle Linux), Nginx (Reverse Proxy, Caching), CI/CD (GitHub Actions), PgBouncer.
- **Database & Data Layer Optimization:** PostgreSQL 15+, PostGIS 3.3+ (Spatial Indexing, SRID 4326, GeoJSON Streaming), MySQL / MariaDB (Master-Slave Replication, Query Optimization, 3NF Normalization, Star Schema Views, Indexed Views), Redis (Object Caching).
- **Security & Regulatory Compliance:** GDPR Compliance, PCI DSS Architecture, Saudi Central Bank (SAMA) Standards, SAML 2.0 / Keycloak SSO Federation, Role-Based Access Control (RBAC), Offline Cryptographic Verification.
- **Web Ecosystems & Decoupled Frameworks:** Python 3.12 (FastAPI), modern PHP 8.x (Symfony, Drupal Enterprise, WordPress FSE/Gutenberg, Moodle LMS), TypeScript, React 19, TailwindCSS, Vite.
- **Languages:** English (Fluent / Native Professional Proficiency), French (Basic / Elementary), Arabic (Native).

PROFESSIONAL EXPERIENCE

Lead Full-Stack Engineer & Architect

Contract Engagements | Institutional, NGO & Digital Platforms | *January 2025 - Present*

- **barincairo.com (2026):** Architected a decoupled Web GIS platform utilizing React 19, Python 3.12 FastAPI, and PostgreSQL/PostGIS. Implemented PostGIS GiST spatial indexing (SRID 4326) and sub-second GeoJSON bounding-box queries for dense vector map streaming. Engineered a multi-stage ingestion pipeline with spatial proximity deduplication (ST_DWithin < 15m), achieving 0% entity duplication across scraper harvest runs.

- **Greek Community of Cairo Modernization (2026):** Modernized an enterprise community portal into a block-native WordPress FSE architecture on PHP 8.2. Developed an automated AST serialization parser to transform 4,870 legacy post builder entries into core Gutenberg blocks with zero data loss. Purged database autoload bloat and integrated Redis object caching, reducing TTFB below 200ms with zero cutover downtime.
- **Greek Community of Alexandria Modernization (2026):** Modernized a 15-year WordPress portal into a lightweight Full Site Editing block theme on PHP 8.2 with full GDPR compliance. Engineered an AST shortcode migration pipeline, automated 15 years of legacy newsletter date parsing, and configured Nginx reverse proxy media streaming.
- **North South Consultants Exchange - NSCE (2025):** Modernized an institutional donor transparency and evaluation platform for **UN, GIZ, EU, and AfDB** development programs. Unified multilingual database records into a normalized schema, built dual-tier spatial governorate filtering for Egypt, and optimized MySQL queries to increase organic search visibility to over 20,000 monthly impressions.

Integration Architect & Technical Lead

Meem Development (SEF Academy) | *January 2024 - December 2024*

- **SAML 2.0 & Keycloak SSO Federation:** Architected and deployed enterprise SAML 2.0 Single Sign-On (SSO) via Keycloak across Moodle LMS, a custom student registration web app, and an administrative portal, resolving authentication token failures and unifying identity management.
- **Secure Database Topology:** Configured role-isolated SSH tunneling protocols for distributed engineering teams, enabling secure concurrent database administration on Test and UAT instances without public port exposure.
- **Transactional Relays & Integrations:** Configured authenticated SMTP relays for deterministic course certificate dispatch and validated enterprise Microsoft Teams integration hooks.

Principal Software Architect & Technical Lead

iSoft (IdealSoft Emirates) | Retainer & Government Contracts | *2020 - 2024*

Contracted to deliver ministry-level digital infrastructure, high-concurrency transactional routing engines, and API architectures for federal and municipal public sector clients across Egypt, Saudi Arabia, and the UAE.

- **National Vehicle Inspection Scheduling (SASO - Saudi Ministry of Standards, Metrology and Quality):** Architected the nationwide vehicle inspection booking system managing multi-lane facilities. Engineered a distributed transactional state-locking engine with a deterministic 5-minute buffer that eliminated race conditions across competing web, mobile, and backend channels for 15-minute appointment slots. Remodeled database schemas into 3NF for transactional safety and Star Schema for real-time reporting. Decoupled Drupal core to expose versioned REST APIs to a high-performance React client.
- **Air-Gapped Intranet Certification Platform (Egyptian Ministry of Interior - Traffic Police Dept.):** Architected and deployed an enterprise Moodle training portal on a restricted state intranet for mandatory licensing courses nationwide with zero internet access. Engineered a deterministic zero-network deployment pipeline using a custom-patched Oracle Linux OS repository on physical USB storage with fault-tolerant Bash migration scripts and offline cryptographic fee verification.
- **National Vehicle Inspection Digital Ecosystem (Aman Portal - Egyptian Ministry of Interior):** Designed the digital booking and dispatch engine for nationwide vehicle inspections, redirecting citizen volume from congested municipal police departments to commercial hubs and mobile inspection vans via high-availability backend API integration.
- **Dubai Police Academy LMS Upgrade (via iSoft):** Modernized and upgraded the mission-critical training LMS for law enforcement cadet curricula. Refactored hardcoded legacy modifications into modular hook-driven plugins, verified multi-step MySQL schema migrations preserving 100% of historical cadet training audit records, and vetoed an unviable Windows migration to mandate a stable enterprise Linux baseline.
- **State-Licensed Cooperative Insurance Platform (Salama Portal - Saudi Arabia):** Engineered core registration workflows, pre-ingestion validation mechanics, and secure transaction-handling layers for compliant digital automotive insurance under Saudi Central Bank (SAMA) regulatory standards.
- **Enterprise Governance & Deployment Reliability:** Enforced a strict 4-tier environment topology (Dev, Testing, Staging, Preproduction) and a structured production deployment schedule, eliminating critical weekend outages.

Lead Full-Stack Engineer & Enterprise Architect

NileTech & Technology Retainers | *2016 - 2019*

- **Enterprise Cloud Infrastructure Migration (AHCC - 2019):** Architected and executed a zero-downtime enterprise cloud migration to AWS using Terraform, ECS, Aurora PostgreSQL, and PgBouncer. Synchronized an active 550 GB database cluster without data loss via asynchronous logical replication and row-level checksum verification. Deployed PgBouncer transaction pooling to absorb bursts of 4,500 concurrent connections, and authored modular Terraform manifests to provision VPCs, multi-AZ subnets, and ECS task definitions in version control.
- **High-Concurrency Digital News Network (Ajel.sa | 2017 - 2019):** Architected and scaled Saudi Arabia's premier digital newspaper on Drupal Enterprise and Acquia Cloud Enterprise. Engineered a multi-tier caching hierarchy combining Varnish edge caching with Redis persistent object caching, offloading over 85% of dynamic database read queries and preserving sub-50ms response times during breaking news traffic surges. Implemented a 3-tier automated CI/CD release pipeline that eliminated cutover downtime.
- **Civic & Cultural Heritage Platform (Turathuna | 2017):** Architected a bilingual digital platform for the Abu Dhabi Cultural Programs & Heritage Festivals Committee (UAE Government) with broadcast TV archives, festival calendars, and native Arabic RTL typography.
- **Startup MVP Scoping & Architecture Mentorship (Egyptian MCIT / ITI | 2017):** Delivered technical architecture and feasibility workshops for the Information Technology Institute (Ministry of Communications and Information Technology). Trained prospective founders on system blueprinting, database bottleneck identification, and operational technical debt prevention.
- **L3D Emergency Blood Donation & Donor Registry Platform (2016):** Contributed pro-bono architecture for a civic healthcare initiative. Modeled clinical ABO/Rh blood compatibility matrices in normalized MySQL schemas, designed geographic proximity matching logic, and architected automated SMS alert dispatch pipelines.
- **Automotive Insurance Portal Stabilization (Saaed via Softech | 2018):** Stabilized an on-premise CentOS WordPress Multisite insurance portal, backporting and pinning legacy RPM packages to eliminate host crashes and refactoring multi-tenant RBAC permissions.
- **Strangler-Fig News Broadcast Migration (218TV.net | 2016):** Executed a Strangler-Fig pattern migration for high-traffic broadcast news operations into an enterprise Drupal architecture, configuring Varnish reverse-proxy caching sustaining 15,000 active concurrent connections.
- **Hospitality Space Allocation Platform (HSP | Focus Media Group | 2017):** Architected a 3-tier event booking, meeting room allocation, and real-time hallway digital signage wayfinding system deployed across luxury hotels (Hilton) and corporate venues.

NGO & Civic Infrastructure Consultant

Orthodox Mission in Botswana | *2010*

- **Low-Bandwidth Regional Community Platform:** Engineered a lightweight community publishing platform for an NGO operating across Southern Africa. Designed normalized relational schemas and optimized page DOM footprints to keep payloads strictly under 500 KB, ensuring accessible load performance across regional 2G/3G low-bandwidth cellular networks.

Co-Founder & Chief Technology Officer (CTO)

Wallety (Fintech Payment Gateway) | *December 2009 - July 2015*

Architected, launched, and scaled an early online payment gateway in Egypt, processing 33% of all domestic air travel ticketing volume.

- **PCI DSS Compliant Core Processing Engine:** Solely engineered the core payment gateway in PHP and Symfony adhering to PCI DSS security standards, handling online payment settlements under unstable telecommunications infrastructure.
- **GDS & Banking Core Integrations:** Built direct integrations with domestic acquiring banks, 200+ enterprise merchants, and the Amadeus Global Distribution System (GDS) for atomic flight seat reservations.
- **Master-Slave Replication & CLI Daemons:** Clustered MySQL separating master ACID writes from slave read-reporting, and engineered 1-second CLI background daemon loops to process transaction batches asynchronously.

Lead Software Engineer & Web Architect

MITCHDesigns | *2009 - 2014*

- **Custom MVC Framework & Enterprise Data Modeling:** Architected a modular PHP MVC framework that served as the technical foundation for over 15 commercial platforms, delivering sub-100ms response times. Designed normalized relational schemas and compound MySQL indexing strategies.
- **Dynamic Contract Pricing & Booking Engines:** Engineered dynamic pricing rules and reservation engines for travel tech platforms including Smash Holidays and Agazat Masr (Tourico). Established agency Git workflows, requirements elicitation SOPs, and developer training.

EDUCATION & FOUNDATIONS

- **Undergraduate Studies in Computer Engineering (Completed 5-Year Curriculum Coursework):** Arab Academy for Science, Technology and Maritime Transport (AASTMT)
  Core Coursework: Operating Systems Design, Relational Database Theory, Distributed Systems, Algorithms & Data Structures, SCADA Systems.
- **Continuous Specialization:** Low-Level LLM Architecture (*AI Engineering from Scratch*), Agentic CLI Architecture (Antigravity), Open-Source Data Pipelines (`php-jobspy`, `freeworld-job-finder`).